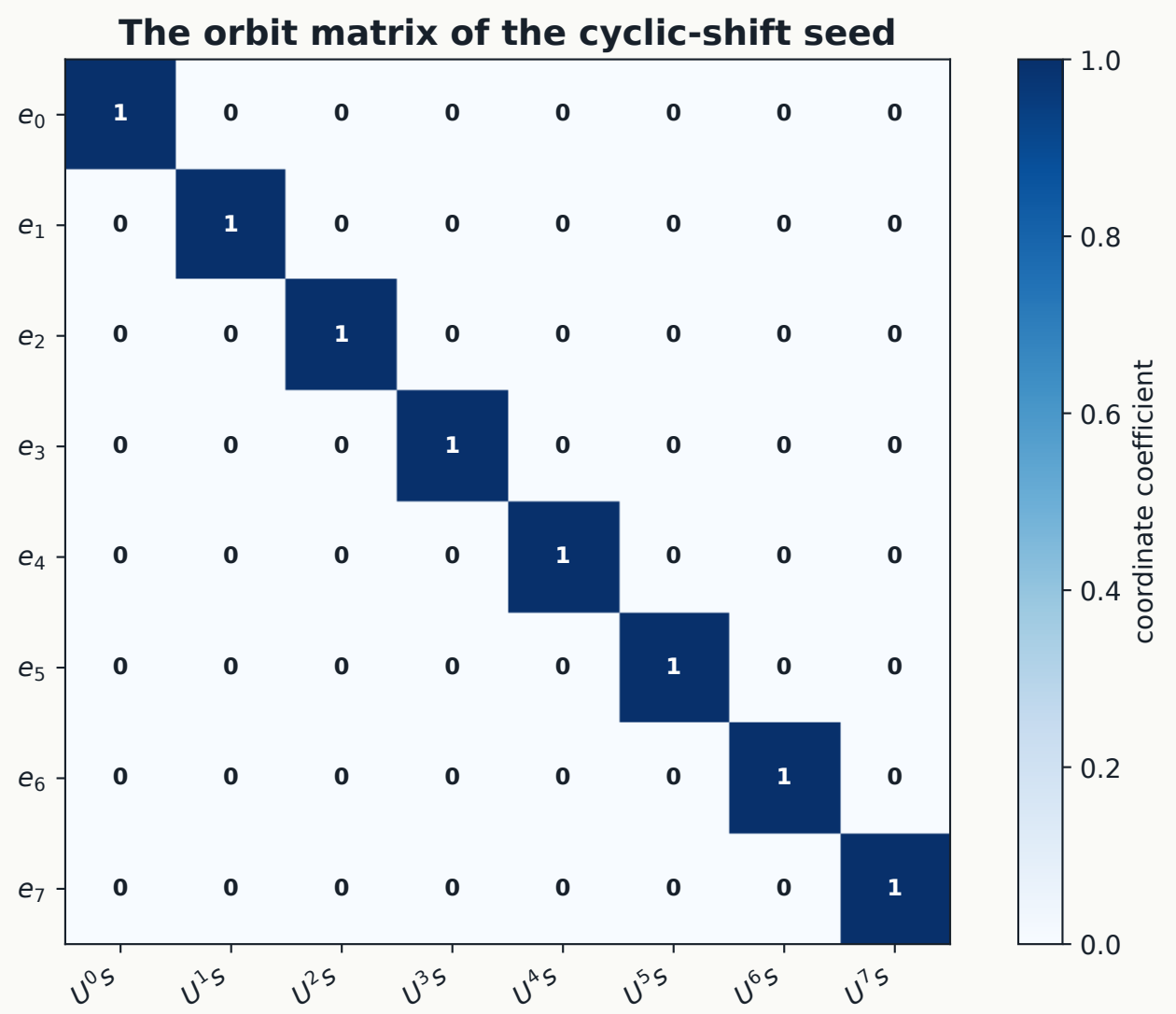

Every basis coordinate appears exactly once, so the orbit rank is 8.